

Yulong Zhao

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Education

University of Arizona — PhD in Statistics	2023 – Expected 05/2028
University of Wisconsin–Madison — MS in Statistics and Data Science	2022 – 2023
University of Science and Technology of China — BS in Mathematics and Applied Mathematics	2018 – 2022

Research Experience

Real-Time Early Warning with a Controlled False-Alarm Rate 2026 – Present

University of Arizona | Advisor: Prof. Xueying Tang

- Built a streaming risk-scoring system that reads each user's partial action sequence and decides when to flag users heading toward failure, with alert thresholds calibrated on the deployed trigger rule so false alarms on users who would succeed stay within a preset budget ($\leq 5\%$).
- Benchmarked ~ 370 model and alerting-policy configurations (GRU, GRU + attention, temporal CNN, causal Transformer, leakage-free ridge-logistic baselines) on 3 interactive tasks under one shared evaluator.
- Best temporal-CNN model caught **98.9%** of failing users (worst task 97.6%) at precision ≥ 0.98 , keeping false alarms $\leq 4\%$ on every task; GRU reached 85.6%, and the logistic baseline broke the budget.

Time-Aware Modeling of User Event Sequences Sept 2023 – Present

University of Arizona | Advisor: Prof. Xueying Tang

- Developed an interpretable model for timestamped event logs that tests, for each action, whether its effect on the outcome is absent, constant, or depends on *when* it happens (functional regression; B-splines + adaptive group LASSO).
- On 19,089 events from 2,668 users, raised cross-validated accuracy from 93.5% to **96.1%** and **halved log loss** ($0.237 \rightarrow 0.118$) vs. a count-feature logistic baseline, winning on all 10 folds.
- Traced the gain to actions whose effect flips sign with timing (early exploration vs. final choice), a signal count features cannot see; quantified uncertainty with 300 bootstrap refits and 1,200 simulations.

Latent User-Segment and Hidden-State Modeling 2026 – Present

University of Arizona | Advisor: Prof. Xueying Tang

- Designed sparse latent-variable sequence models (hidden Markov model + persistent binary user traits) to identify which traits drive action choice, interface choice, and state transitions; applied to 1,340 users of a multi-app (email + spreadsheet) task.
- Implemented a custom forward-backward EM in Python (multi-start, sparse-group penalty screening, unpenalized refit, BIC selection over up to 64 structures); ran 580+ Slurm jobs with automated fit diagnostics.
- Profiled and optimized the EM core with NumPy/Numba for **up to $3.7\times$ speedup**, with bitwise-identical parameters and objective traces.

Time-Series Forecasting and Regime-Shift Detection May 2024 – Present

University of Arizona | IMSI Interdisciplinary Research Cluster

- Built a walk-forward forecasting pipeline (RBF-SVR, time-ordered CV, no look-ahead) on PCA-compressed features from 8 environmental series; beat the historical-mean baseline in **18/18** settings with **38% lower MSE** on average.
- Developed a regime-shift detector across 28 series pairs; tested significance against 4,999 autocorrelation-preserving surrogates with Holm correction, and confirmed a false-positive rate $\leq 2/50$ by simulation.
- Stress-tested findings under 49 data perturbations, which flagged 1 of 6 candidate shifts as fragile; delivered visual reports to geoscience collaborators.

Publications and Presentations

Zhao, Y., & Tang, X. Scalar-on-process model for educational assessment: modeling time-varying action effects with group LASSO regularization. *Under review at Psychometrika*.

Refereed oral presentation of the above, Symposium on Data Science & Statistics (SDSS), Milwaukee, WI, April 2026.

Skills

Programming: Python (NumPy, pandas, scikit-learn, Numba), R (Keras/TensorFlow), SQL, Git, Slurm/HPC, LaTeX.

Machine learning: sequence models (GRU, TCN, Transformer), regularized GLMs, kernel methods, latent-variable models/EM.

Statistics: threshold calibration, leakage-safe validation, bootstrap, multiple testing, simulation studies.

Awards

Research Fellowship (2024–26) & Incoming PhD Award, U. Arizona; Academic Excellence Award, UW–Madison; Outstanding Undergraduate Award, USTC.